



39th International Conference
On VLSI Design



25th International Conference
On Embedded Systems

3rd - 7th January 2026 | JW Marriott | Pune, India

Global Synergy in Silicon: VLSI and Embedded AI for Sustainable Computing and Next-Gen Electrified Mobility

TITLE:

Autonomous RTL Verification and Testing of RISC-V Processors using Agentic AI

A novel AI-driven methodology to automate and accelerate RTL functional verification

CHALLENGES:

- Current methods rely heavily on extensive simulation, which increases manufacturing and testing costs, yet still fails to capture all functional bugs and rare corner cases.
- Formal verification provides mathematical correctness but requires significant manual effort to define properties, making it impractical at scale.
- The RISC-V ecosystem is rapidly expanding, creating a need for scalable, intelligent, and cost-effective verification techniques.

MOTIVATION:

- Reduce Manual Workload:** Current verification relies on slow, manual processes and brute-force simulation, which increases costs and still fails to find all critical bugs and corner cases.
- Democratize Advanced Tools:** By using open-source tools, we can make powerful verification techniques accessible to students and researchers who lack access to expensive industrial software.
- Blend AI Power with Domain Expertise:** Our core motivation is to create highly specialized and efficient AI agents. We aim to achieve this by fine-tuning **Small Language Models (SLMs)** with deep, domain-specific knowledge of the RISC-V architecture and RTL verification rules, creating a powerful yet accessible solution.
- Enable a Scalable Future:** The goal is a proof-of-concept for an intelligent and automated verification flow that can scale with the growing complexity of RISC-V designs, ultimately improving design correctness and reducing manual effort.

PROPOSED SOLUTION:

Methodology: Combine multi-agent AI systems with RTL design and simulation to validate a RISC-V CPU core.

Goal: Make advanced verification workflows accessible, intelligent, and low-cost, especially for students and researchers.

Key Components:

RTL Design & Simulation: Establish baseline environment for testing and debugging using standard HDL descriptions.

AI-Driven Multi-Agent Framework:

Case Generation Agent → Produces meaningful testbenches (pipeline hazards, memory conflicts, etc.)

Coverage Analysis Agent → Tracks code & functional coverage to identify unverified areas.

Missing Case Suggestion Agent → Recommends rare or corner-case scenarios for better coverage.

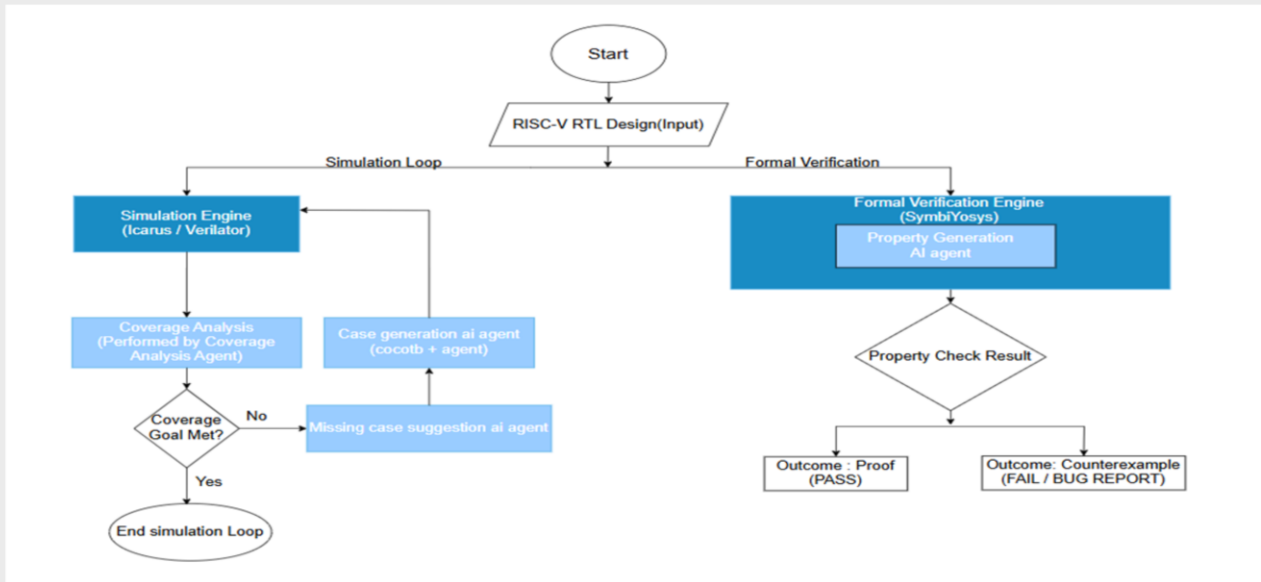
Formal Verification Integration: Property Generation AI agent converts design specs into formal properties to mathematically verify critical components (e.g., ALU, register file) .

Case Generation Agent- Automatically creates testbenches and smart, constrained-random test cases. Instead of relying on brute force, it uses its understanding of the RISC-V ISA to generate meaningful scenarios like pipeline hazards, memory access conflicts, or even division-by-zero situations.

Coverage Analysis Agent- During test execution, this agent monitors both code coverage (lines, branches) and functional coverage (microarchitectural states exercised).

Missing Case Suggestion Agent- Using the coverage reports this agent proposes new high-value test cases to target untested areas.

AGENTIC AI WORKFLOW:



The Simulation

- LoopOn the left is the core of our intelligent framework—the simulation feedback loop. It begins by running tests on the design in a simulation engine like Verilator or Icarus Verilog.
- The Coverage Analysis Agent then examines the results to find any untested areas of the design. A decision is then made: if the coverage goal has been met, this iterative process terminates.
- If not, the Missing Case Suggestion Agent proposes new, high-value test cases to target those specific gaps.
- Finally, the Test Case Generation Agent creates these smart, constrained-random tests, which are then fed back into the simulation engine to start a new, more effective cycle.

The Formal Verification Path

- On the right, we have the formal verification path, which provides a complementary layer of assurance. This path uses a tool like SymbiYosys to mathematically prove that critical components, such as the ALU or register file, adhere to their specifications.
- A Property Generation AI Agent assists this process by helping to convert design specs into formal properties. This concludes with one of two definitive outcomes:

either a 'Proof' of correctness, giving us mathematical certainty, or a 'Counterexample', which is a precise bug report showing exactly how a property failed.

NOVELTY & KEY INNOVATIONS:

Shifting from Brute-Force to Intelligent Verification

- Multi-Agent System:** The first known application of a multi-agent AI system specifically for open-source RTL verification.
- Domain-Aware AI via SLMs:** By fine-tuning Small Language Models (SLMs) with domain-specific data (RISC-V ISA, Verilog). This creates expert agents that are more efficient and accurate for verification tasks than generic models.
- Automated Test & Property Generation:** Moves beyond simple random testing to generate context-aware stimulus and assists in the automatic creation of formal properties.
- Fully Open-Source & Accessible:** Democratizes advanced verification by building a powerful workflow using freely available tools.
- Scalable Proof-of-Concept:** Establishes a methodology that can scale from small student projects to more complex SoC designs.

EVIDENCE & EXPECTED RESULTS:

Projected Verification Improvements

- Smarter Test Generation:** AI is expected to generate tests that cover complex corner cases like pipeline hazards, memory access conflicts, and arithmetic overflows.
- Faster Coverage Closure:** Achieve high code and functional coverage in significantly less simulation time compared to manual or purely random methods. Approximately 25-30% speed up in simulation cycles, 80-85% incremental functional coverage as compared to approximately 60-65% coverage through manual/random methods.
- Automated Bug Hunting:** The system is projected to automatically identify and suggest missing test scenarios targeting rare instruction sequences or edge timing conditions.
- Proven Correctness:** Formal verification integration will provide mathematical assurance for critical components like the ALU and register file.

APPLICATIONS & IMPACT:

•Academic & Research Impact:

- Empowers Innovation:** Provides a powerful, open-source tool for students and researchers to validate novel RISC-V architectures without the barrier of expensive commercial licenses.
- Modernizes Education:** Serves as a hands-on educational platform for teaching next-generation, AI-driven VLSI verification concepts, better preparing students for industry challenges.

•Industry & Open-Source Impact:

- Accelerates Startups:** Offers a low-cost, scalable pre-silicon validation solution for startups and SMEs, reducing time-to-market and verification costs.
- Paves the Way for Future EDA:** Establishes a foundational proof-of-concept for integrating autonomous AI agents into large-scale industrial verification and debug workflows.

FEASIBILITY ROADMAP

•Projected Outcomes:

We are currently evaluating a basic implementation of our idea on PicoRV32 core. Early results indicate that our proposed methodology works and is scalable.

•Proof-of-Concept: A functional end-to-end pipeline demonstrating the agentic AI workflow on a PicoRV32 core.

•Feasibility Roadmap:

- Productise the base simulation setup and design of AI agents.
- Domain specific SLM training and closed loop pipeline implementation.
- Quality analysis and collection of metrics from detailed experiments to prepare the final demo.

This slide outlines our projected results and the concrete plan to achieve a working proof-of-concept.

Our roadmap is structured in three phases, progressing from setting up the baseline simulation to implementing the full closed-loop AI agent pipeline.

The final is dedicated to running experiments to gather the data that will support our projected outcomes on coverage and efficiency gains.

While our immediate goal is a robust simulation-based demonstration, the long-term vision is to map this AI framework into silicon as a "Verification Assist Engine," directly enhancing on-chip debug and validation capabilities.

FUTURE WORK:

SoC-Level Verification: Extend from single cores to full SoCs with interconnects & peripherals.

Reinforcement Learning: Adaptive test generation using historical bug data.

Hardware Acceleration: On-chip verification with FPGA-based AI accelerators.

CONCLUSION & ACKNOWLEDGEMENTS:

- Agentic AI presents a promising new frontier for automating and accelerating RTL verification.
- Enables scalable and intelligent RTL verification, reducing manual effort and improving design correctness for future RISC-V processors.
- This research leverages several powerful open-source projects, including Verilator, Icarus Verilog, CrewAI, Gemma, Cocotb, SymbiYosys, LangChain, and the PicoRV32 core. We gratefully acknowledge the contributions of their respective communities.

This project serves as a foundational proof-of-concept. Our future work will focus on scaling this methodology to handle more complex, multi-core SoC designs. We plan to explore more advanced AI techniques, such as using LLMs for spec-to-property generation and reinforcement learning for adaptive test strategies. The ultimate goal is to prototype this system on an FPGA, demonstrating a true hardware-accelerated, AI-driven verification platform.